

```

import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.decomposition import PCA
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
import matplotlib.pyplot as plt

# -----
# Load Dataset
# -----
df = pd.read_csv('/content/Leukemia_GSE9476.csv')

# -----
# Check columns (IMPORTANT)
# -----
# -----
# Handle Target Column
# -----
# Replace 'type' with your actual target column name if different
target_column = 'type' # CHANGE if needed

y = df[target_column]
X = df.drop(columns=[target_column])

# -----
# Encode Target (ALL / AML → 0/1)
# -----
le = LabelEncoder()
y = le.fit_transform(y)

# -----
# Train-Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# -----
# Feature Scaling
# -----
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# -----
# PCA (VERY IMPORTANT)
# -----
pca = PCA(n_components=10)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

# -----
# Model Training
# -----
model = LogisticRegression(max_iter=1000)
model.fit(X_train_pca, y_train)

# -----
# Prediction
# -----
y_pred = model.predict(X_test_pca)

# -----
# Evaluation
# -----
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

# -----
# Visualization (PCA)
# -----
plt.scatter(X_train_pca[:, 0], X_train_pca[:, 1], c=y_train)
plt.xlabel("PCA Component 1")
plt.ylabel("PCA Component 2")
plt.title("Microarray Data (PCA Visualization)")
plt.show()

```

Accuracy: 0.9230769230769231

Classification Report:

	precision	recall	f1-score	support
0	1.00	0.67	0.80	3
1	1.00	1.00	1.00	3
2	1.00	1.00	1.00	2
3	1.00	1.00	1.00	2
4	0.75	1.00	0.86	3
accuracy			0.92	13
macro avg	0.95	0.93	0.93	13
weighted avg	0.94	0.92	0.92	13

